

OpenSTA Important Commands Cheat Sheet

Basic

sta - Launch OpenSTA
sta -help
sta -version
exit

Design Loading

read_liberty osu018_stdcells.lib
read_verilog design_synth.v
link_design top_module
read_sdc design.sdc

Timing Reports

report_checks
report_checks -path_delay max
report_checks -path_delay min
report_checks -sort_by_slack
report_worst_slack -max
report_worst_slack -min
report_check_types -violators

Clock & Constraints

create_clock -name clk **-period** 10 [get_ports clk]
set_input_delay -clock clk **-max** 5 [get_ports in]
set_output_delay -clock clk **-max** 2 [get_ports out]
set_load -pin_load 4 [get_ports out]
set_input_transition -max 5 [get_ports in]
set_clock_latency 1 [get_clocks clk]
set_clock_transition 0.2 [get_clocks clk]

Utilities

report_clock_skew
report_power
report_units
report_instance U1
report_cell U1
get_ports *
get_cells *
get_pins *
get_clocks *

Multi-Corner

define_corners wc typ bc
read_liberty -corner wc slow.lib
read_liberty -corner typ typical.lib
read_liberty -corner bc fast.lib
set_timing_derate -early 0.9
set_timing_derate -late 1.1

Typical Flow

```
sta
read_liberty osu018_stdcells.lib
read_verilog design_synth.v
link_design top_module
read_sdc design.sdc
report_checks
report_checks -path_delay max
report_checks -path_delay min
exit
```